

Software Engineering Project Jan-2026

Milestone 2 Deliverable Report



Sports Facility Booking & Club Management System

Submitted by,

Team Code : Team-096

Team Name: Agile_Alliance

**IITM Online BS Degree Program, Indian Institute of Technology, Madras, Chennai,
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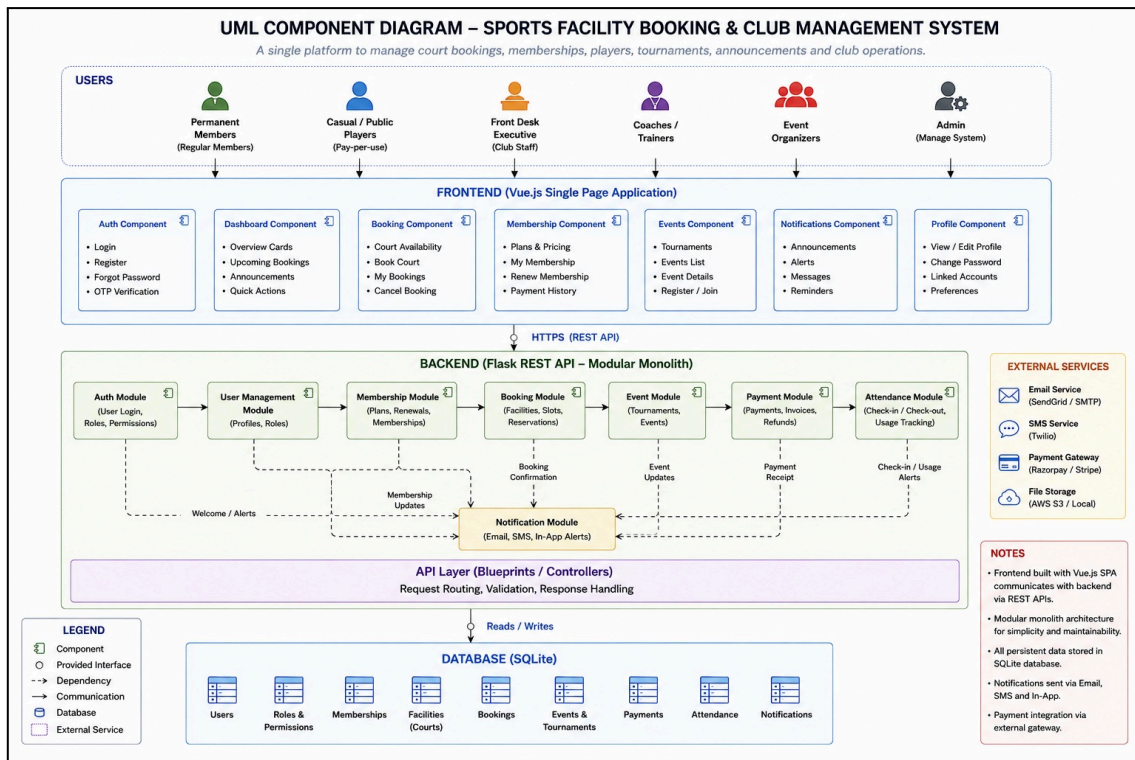
PROJECT DETAILS:

1. **Domain** : Community Service Platform
2. **Sub-domain** : Sport Club Management
3. **Project Title** : Sports Facility Booking & Club Management
4. **Date of Submission** : 23.07.2026

TEAM MEMBER DETAILS:

Roll No.	Name	Role
21f30002906	Harshita Miranda	Scrum Master, Backend Developer, Product Manager
22f1001445	Abdul Hadi	Scrum Master, Backend Developer, Frontend Developer, Project Manager.
23f1003046	Gajula Srinivas Jatinkrishna	Code Reviewer, Backend Developer
24f1001895	Jayesh Bansal	Tester, Backend Developer
23f2001305	Varun Karthik B	Product Manager, Frontend Developer

DESIGN OF COMPONENTS



1. User & Player Component (Dashboard & Profile APIs)

- Access role-specific dashboards with overview cards, upcoming bookings, and announcements based on user type (Permanent, Casual, Front Desk, Coach, etc.).
- Update personal profiles, manage linked accounts, and configure communication preferences.
- Securely change account passwords and manage personal details directly from the frontend.
- View general announcements and execute quick actions immediately upon logging in.

2. Booking Management Component (Facility & Reservation APIs)

- View real-time court and facility availability based on preferred sport types and time slots.
- Submit structured booking requests for specific facilities and receive instant digital confirmations.
- View a comprehensive list of all past and upcoming personal bookings.
- Cancel active reservations seamlessly and trigger automatic schedule updates.
- Ensure no double-booking through backend validation and slot management.

3. Membership Management Component (Plans & Subscriptions APIs)

- Browse available membership plans, including detailed pricing, duration, and associated benefits.
- Subscribe to new memberships or upgrade existing ones directly through the user portal.
- Manage auto-renewal preferences and track membership expiration dates.
- Access a complete history of membership-related payments and invoices.
- Allow the backend to verify active membership status automatically before applying discounts or allowing exclusive bookings.

4. Event & Tournament Component (Organization APIs)

- Create and manage sports tournaments, matches, and special events (accessible by Event Organizers and Admins).
- View a detailed list of upcoming events, including schedules, descriptions, and registration fees.
- Register or join events directly through the platform as a player.
- Push automated event updates and schedule changes to registered participants.
- Track attendee limits and manage ongoing tournament brackets or schedules.

5. Attendance & Usage Tracking Component (Operations APIs)

- Record precise check-in and check-out timestamps for users arriving at the facility.
- Monitor live facility usage and track active hours across different courts.
- Send automated usage alerts to staff for capacity management or security purposes.
- Link attendance records directly to specific bookings or event registrations for audit purposes.

6. Facility Administration Component (Backend APIs)

- Define and manage the physical facilities (e.g., tennis courts, swimming pools) with assigned sport types and unique hourly rates.
- Update facility status (e.g., active, under maintenance) to automatically reflect in the booking availability.
- Isolate bookings and schedules strictly per facility to maintain clean data flow.
- Generate facility usage statistics based on backend booking and attendance data.

7. Payment & Billing Component (Financial APIs)

- Process secure payments for bookings, memberships, and event registrations.
- Generate digital payment receipts and automated invoices for every successful transaction.
- Track payment status, payment methods, and unique transaction IDs for transparent auditing.
- Handle automated or manual refunds, logging the exact refund amount and reason.
- Synchronize all financial transactions securely with the SQLite database.

8. Authentication & Access Control Component (Auth APIs)

- Register new users with clear role definitions (Permanent Member, Casual Player, Front Desk Executive, Coaches, Admin).
- Securely log in users via standard credentials and manage sessions.
- Verify accounts securely using OTP (One-Time Password) validation during the registration process.
- Handle "Forgot Password" requests safely to ensure account recovery.
- Enforce strict role-based access control (RBAC), ensuring users only access modules permitted by their assigned roles.

9. Notification & Alert Component (Messaging APIs)

- Deliver targeted announcements, system alerts, and direct messages to specific user groups or the entire club.
- Trigger automatic booking confirmations, event reminders, and payment receipts.
- Manage in-app alerts directly on the Vue.js frontend dashboard.
- Track whether notifications have been successfully delivered and read by the recipients.
- Route automated system alerts to staff for facility monitoring and attendance updates.

10. External Services Integration Component (Third-Party APIs)

- Integrate with external Email Services (e.g., SendGrid/SMTP) for reliable delivery of official communications.
- Connect with SMS Services (e.g., Twilio) for instant mobile alerts and OTP delivery.
- Route all financial transactions through trusted external Payment Gateways (e.g., Razorpay/Stripe).
- Manage static assets and user uploads by linking to File Storage solutions (e.g., AWS S3).

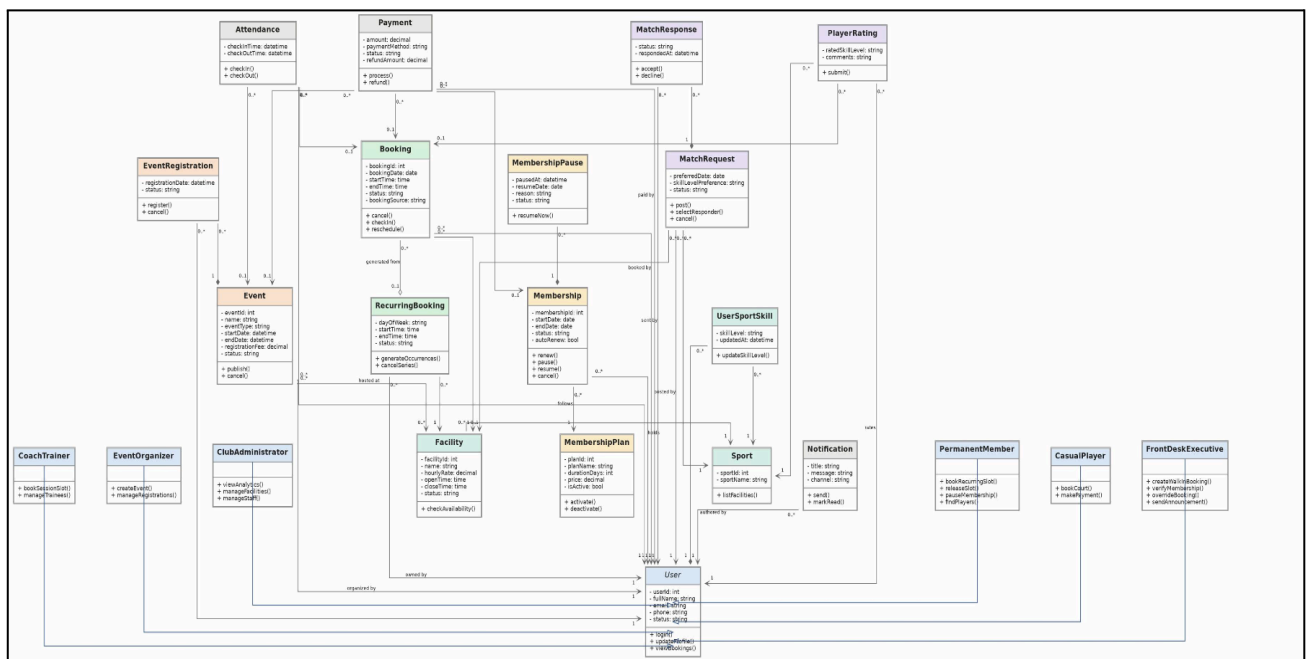
SYSTEM DESIGN - CLASS DIAGRAM

Overview

The class diagram represents the complete system architecture of a Sports Club Management System consisting of numerous interconnected classes organized across several functional modules. It illustrates the relationships between core components such as user and member management, event and facility scheduling, match organization, skill ratings, and payment operations, ensuring a cohesive and functional design.

Key Highlights

- Covers major modules including User & Member Management, Event & Facility Bookings, Matchmaking, Skill Rating, and Payments.
- Clearly defines relationships between entities such as User, Event, Booking, Facility, Membership, MatchRequest, and Payment.
- Supports common club workflows including member registration, facility reservation, event management, player matchmaking, skill assessment, and attendance tracking.
- Designed for scalability and modular development.



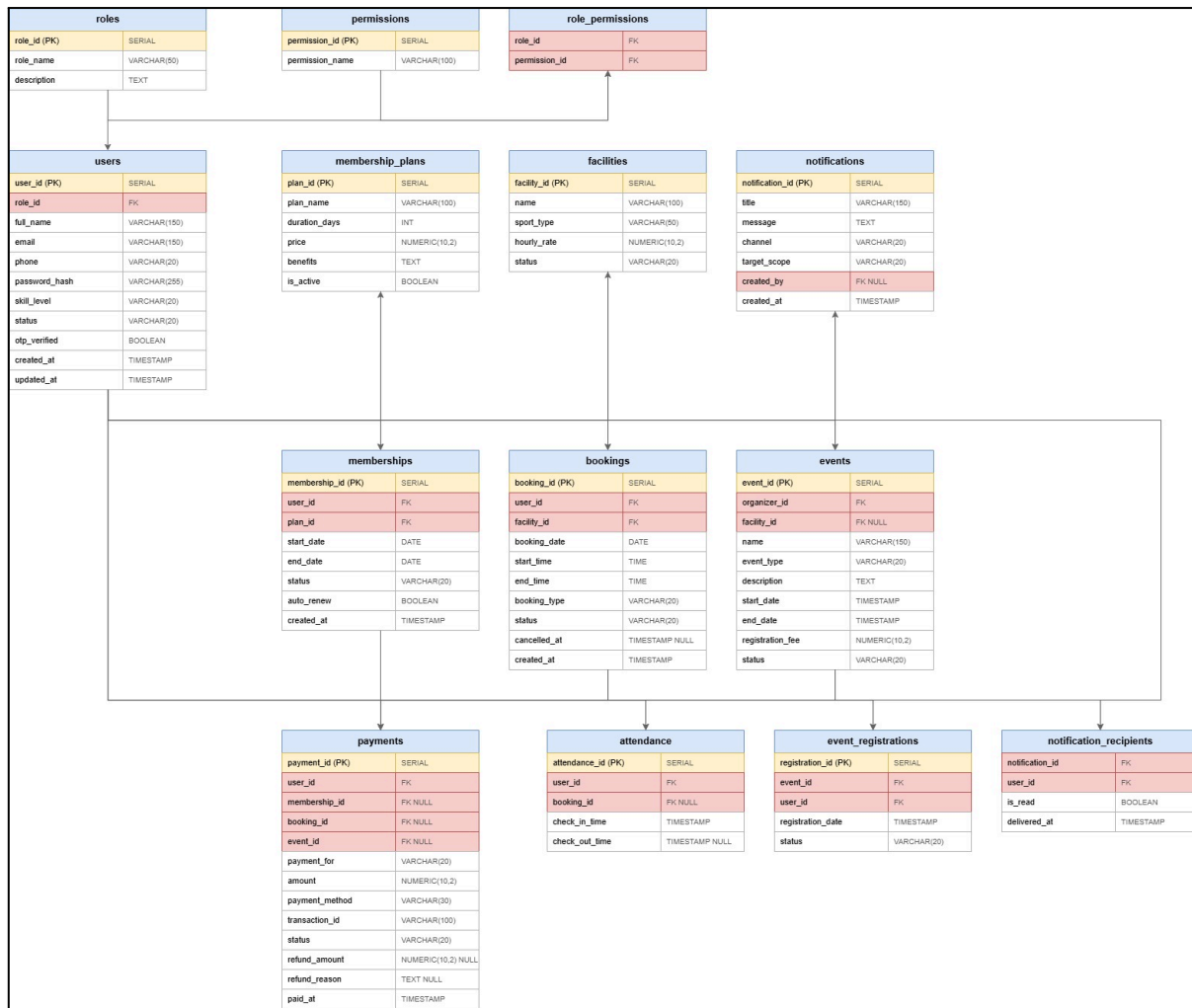
DATABASE DESIGN – ER DIAGRAM

Overview

The ER (Entity-Relationship) diagram represents the database structure of the Sports Club Management System, defining entities, attributes, and relationships required for efficient data management. It captures how different components such as users, memberships, facility bookings, events, and payments are interconnected at the database level.

Key Highlights

- Defines relationships such as:
 - One-to-Many (Roles → Users, Facilities → Bookings, Users → Payments)
 - Many-to-Many resolutions (Users & Events via Event Registrations, Roles & Permissions via Role Permissions)
- Ensures data integrity and consistency across all modules.
- Supports core club operations including membership management, facility scheduling, event coordination, and attendance tracking.
- Optimized for efficient querying and scalability.



SPRINT SCHEDULE

Sprint 1: Milestone 1 – Identify User Requirements

- Identify users of the application (primary, secondary, tertiary)
- Conduct interviews/observations to identify pain points (no direct feature suggestion)
- Record proof (audio/video snippets)
- Write SMART user stories: As a [user], I want [action], so that [benefit]
- Prepare deliverables (PDF: users, research, pain points, user stories)
- **Date: (Before 28/06/2026)**

Sprint 2: Milestone 2 – Planning & Design

- Finalize project branding, tech stack, and tools
- Create full project schedule + Gantt chart
- Setup Jira/Trello board and tasks
- Design system architecture (components, class diagram)
- Create database schema (ER diagram)
- Develop and link major UI screens
- Prepare Milestone 2 PDF (diagrams, UI, schedule, screenshots)
- **Date: 28/07/2026 - 22/07/2026**

Sprint 3: Milestone 3 – API Development (Sprint 1)

- Identify and prioritize core APIs (6–8 endpoints)
- Develop APIs with proper request/response structure (YAML)
- Implement validation and error handling
- Map APIs to user stories
- Perform basic testing using pytest
- Track bugs/issues using GitHub
- **Date: 22/07/2026 - 02/08/2026**

Sprint 4: Milestone 4 – Integration & Testing (Sprint 2)

- Integrate frontend with backend APIs
- Implement core user flows
- Design and execute test cases (input/expected/actual)
- Fix integration and functional bugs
- Document test results and feedback
- Update README and project documentation
- **Date: 02/07/2026 - 12/08/2026**

Sprint 5: Milestone 5 – Final Testing & Report Preparation

- Demo complete working application
- Run final pytest and capture results
- Document failures and fixes
- Collect and summarize user feedback
- Compile final PDF report (all milestones + screenshots)
- Prepare slides and draft video demo
- **Date: 19/08/2026**

Sprint 6: Final Submission

- Final sanity check of project (code + documentation)
- Finalize video recording and upload
- Complete report proofreading and run instructions
- Prepare final code zip + README
- Submit all deliverables (PDF, code, video, slides)
- **Date: 23/08/2026**

SCRUM MEETINGS SCHEDULE AND MINUTES

Mandatory SCRUM Meetings were conducted via Google Meet as weekly sessions held on Wednesdays at 10.00 P.M apart from the client calls and milestone submission discussions.

Meeting Minutes from June 18, 2026 held at 10.00 P.M. - 11.00 P.M. :

- Team introductions and discussion on individual roles and expertise
- Analyzed multiple project domains and finalized on the domain “Community Service Platform”
- Brainstormed the plan of action for milestone one and project tools to be used such as Github projects for seamless task assigning using kanban boards.

- The team decided to split the work for Milestone 1 in 3 parts :
 - Interview Questionnaires preparation & User Identification
 - User Interviews and pain points gathering
 - User story building

Meeting Minutes from June 26, 2026 held at 10.00 P.M. :

- Clarified individual responsibilities and discussed the feedback given by the client.
- Reviewed the team progress:
 - Identified the primary, secondary and tertiary users
 - Analyzed interview data and documented key pain points.
- Discussed the Github setup plan & the deliverables required for milestone 1.

Meeting Minutes from June 28, 2026 held at 10.00 P.M. :

- Conducted final drafting and verification of all Milestone 1 deliverables.
- Completed sanity check of documentation, transcripts, and user stories.

Meeting Minutes from July 8, 2026 held at 10.00 P.M. :

- A Walk through of the Github setup and the code conventions to follow were discussed with the whole team.
- Overall Project Schedule and Plan of Action for Milestone 2 was drafted.
- Each team member was assigned a task as per their choice of roles. The tasks were split for milestone 2 as follows:
 - Generation of UI Mockups
 - Creation of Class diagrams
 - Creation of Database diagrams
 - Identifying the Components of the system
 - Frontend code generation & code reviews were assigned to the team members with the respective roles.

Meeting Minutes from July 14, 2026 held at 10.00 P.M. :

- Progress and client feedback discussion.
- UI/UX was reviewed by the whole team
- Demonstrated UI pages with navigation and responsiveness.

Meeting Minutes from July 22, 2026 held at 10.00 P.M. :

- Deliverables for milestone 2 were discussed.
- The pending tasks were compared against the project schedule.
- The deliverable documentation was drafted for the completed sections.
- API specifications were developed and reviewed.

PROJECT SCHEDULING TOOLS

To manage the project effectively, we used visual task management and scheduling tools to organize work across different phases and ensure timely completion of milestones.

Tools Used:

- **Kanban Board in Github Projects** (for task management and tracking)
- **GitHub Issues** (for bug tracking and issue logging)
- **Gantt Chart (Excel)** (for project timeline visualization)

Description:

We primarily used a **Kanban board** to manage and track the progress of tasks throughout the project. Tasks were categorized into columns such as *To Do*, *In Progress*, and *Completed*, allowing the team to clearly visualize the workflow and monitor progress at each stage of development. This improved transparency and ensured that all team members were aware of their responsibilities.

For issue tracking during development and testing, **GitHub Issues** was used to log bugs, blockers, and improvements. Each issue was documented properly and tracked until resolution, which also helped in maintaining a record for reporting and evaluation purposes.

Additionally, a **Gantt chart** was created to represent the complete project schedule from February to April. This provided a high-level view of milestones, deadlines, and dependencies, ensuring that all phases such as design, API development, integration, and testing were completed on time.

Outcome:

These tools helped in:

- Efficient tracking of tasks and progress
- Clear visualization of workflow
- Proper documentation of issues
- Timely completion of all milestones

PROJECT GANTT CHART

Overview

The Gantt chart represents the complete timeline of the project from **Milestone 1 (User Requirement Analysis)** to **Final Submission (April 19, 2026)**. It provides a visual representation of all major phases, including planning, design, development, integration, testing, and final delivery.

The chart illustrates how the project is divided into multiple milestones and sprints, ensuring a structured and systematic approach to development.

Description

The project begins with **Milestone 1**, which focuses on identifying user requirements through interviews, observations, and formulation of SMART user stories. This phase lays the foundation for all subsequent development activities.

Following this, **Milestone 2** covers planning and design activities, including project scheduling, Gantt chart creation, system design (class diagrams and database schema), and UI development. This phase ensures that the architecture and workflow of the system are clearly defined before implementation begins.

The development phase is carried out in **Sprint 1 (Milestone 3)**, where core APIs are designed, implemented, and tested. Tasks are divided into initial API development and completion with testing, ensuring incremental progress.

Next, **Sprint 2 (Milestone 4)** focuses on integrating the frontend and backend components. This phase includes implementing core user flows, conducting detailed testing, fixing bugs, and collecting feedback.

Finally, **Milestone 5** involves final testing, report preparation, video demonstration, and submission of all deliverables.

Key Features of the Gantt Chart

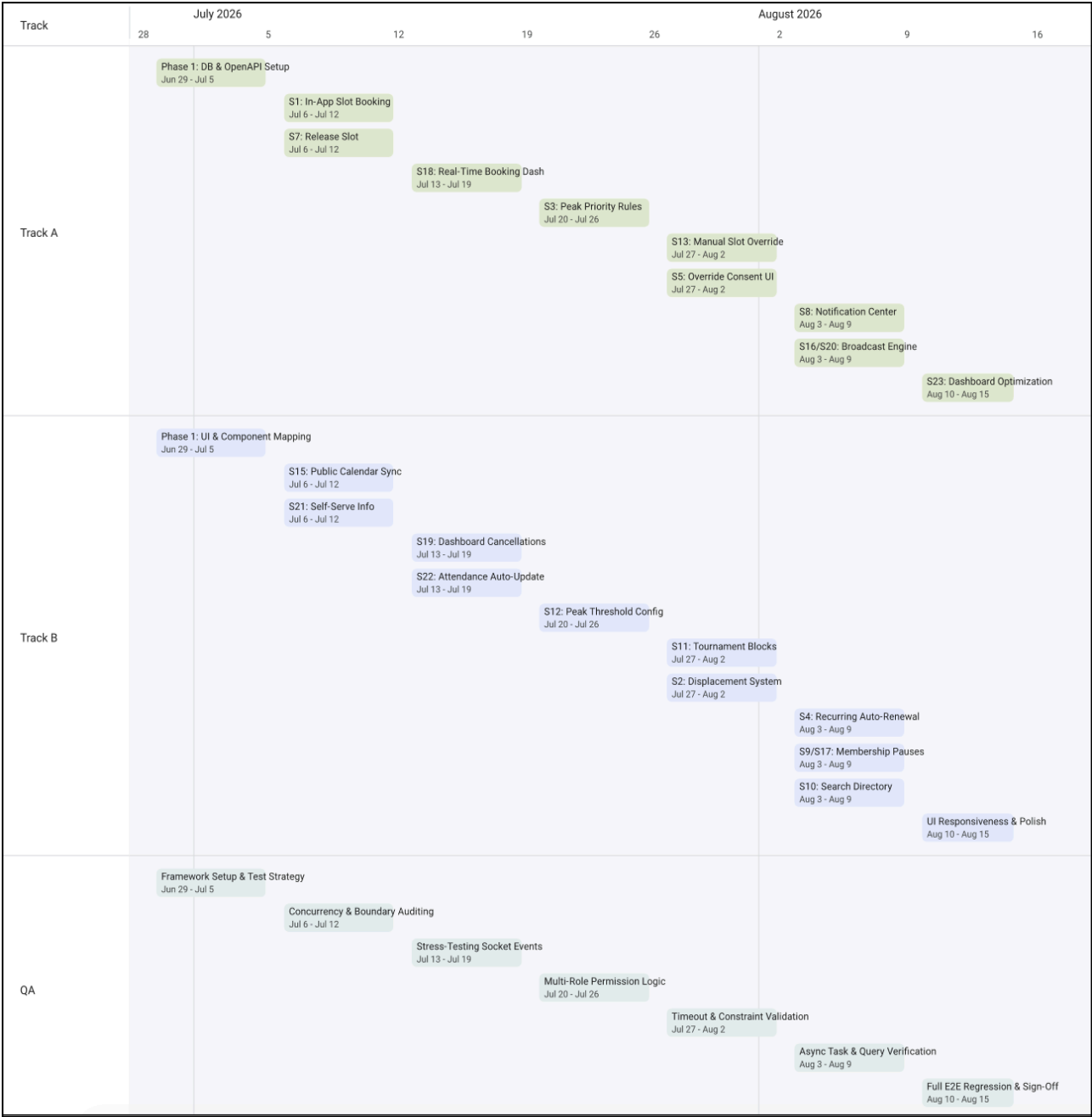
- Clearly shows the **timeline of all milestones and sprints**
- Represents **parallel and overlapping tasks** (e.g., UI development and system design)
- Highlights **dependencies between phases** (design → development → testing → submission)
- Ensures **efficient time management and task tracking**
- Helps in monitoring progress and meeting deadlines

Outcome

The Gantt chart played a crucial role in:

- Planning the project effectively
- Managing task dependencies and overlaps
- Tracking progress across milestones and sprints
- Ensuring timely completion of the project within the deadline

Gantt Chart Task Table



PROJECT TRACKING MECHANISM

Overview

To effectively manage and track project progress, a **Kanban board** was used as the primary project tracking tool. It provided a visual representation of tasks across different stages of development, ensuring transparency and efficient workflow management.

Description

The Kanban board was structured into multiple columns representing task status:

- **Backlog** – Tasks that are planned but not yet started
- **To Do** – Tasks ready to be picked up
- **In Progress** – Tasks currently being worked on
- **In Review** – Tasks under review or testing
- **Corrective Actions** – Tasks which require changes
- **Validation** - Tasks which require manual testing
- **Done** – Completed tasks

Each task was represented as a card containing details such as task title, priority, and assignee. Tasks were continuously moved across columns as progress was made.

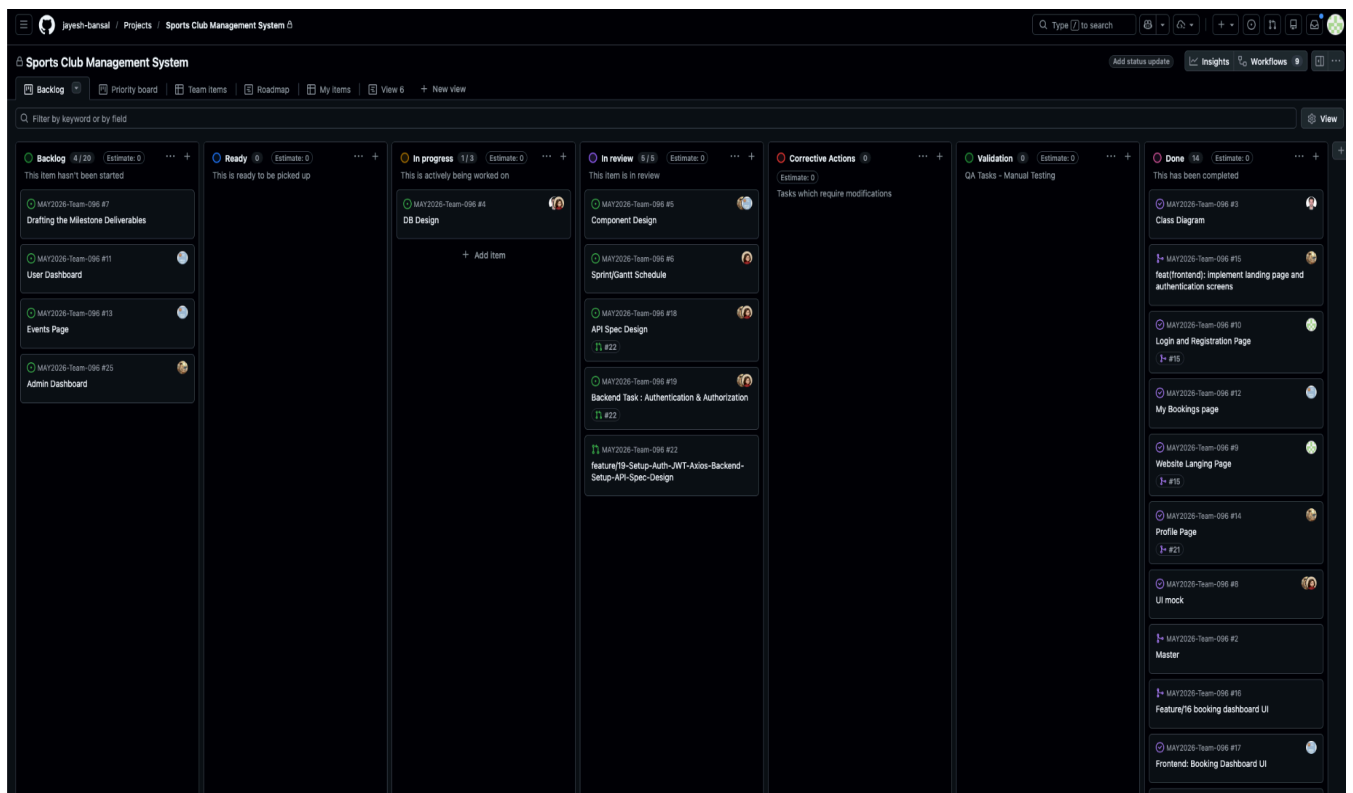
The board also included:

- Task prioritization (P0, P1, etc.)
- Assignment of responsibilities to team members
- Real-time progress tracking
- Integration with GitHub for issue tracking

Outcome

The Kanban board helped in:

- Efficient task tracking and management
- Clear visualization of project progress
- Better team coordination and accountability
- Timely completion of milestones and deliverables



FRONTEND INTERFACE SCREENS

Overview

The frontend of the **Sports Facility Booking & Club Management System** was developed using **Vue.js 3** with **Vite** as the build tool. The application follows a **mobile-first, responsive design approach** to ensure seamless usability across desktops, tablets, and mobile devices.

To maintain a consistent user experience, a centralized design system (**global.css**) was implemented. The design system standardizes colors, typography, spacing, responsiveness, and interactive elements throughout the application.

Design System

Primary Colors

- Navy (#0F172A)
- Blue (#2563EB)
- Orange (#F97316)

Typography

- Poppins (Headings)
- Inter (Body Text)

Layout

- Mobile-first responsive grid system
- **UI Components**
- Reusable cards, buttons, navigation menus, and status badges

Visual Effects

- Consistent spacing
- Shadows
- Hover animations
- Smooth transitions

The frontend was designed with a focus on usability, responsiveness, and consistency, ensuring a seamless experience across different devices and screen sizes.

1.Landing Page (/)

The Landing Page serves as the entry point to the Sports Facility Booking & Club Management System. It introduces users to the platform, highlights its key features, and provides quick access to authentication pages. The interface follows a clean, modern design with responsive layouts and reusable UI components.

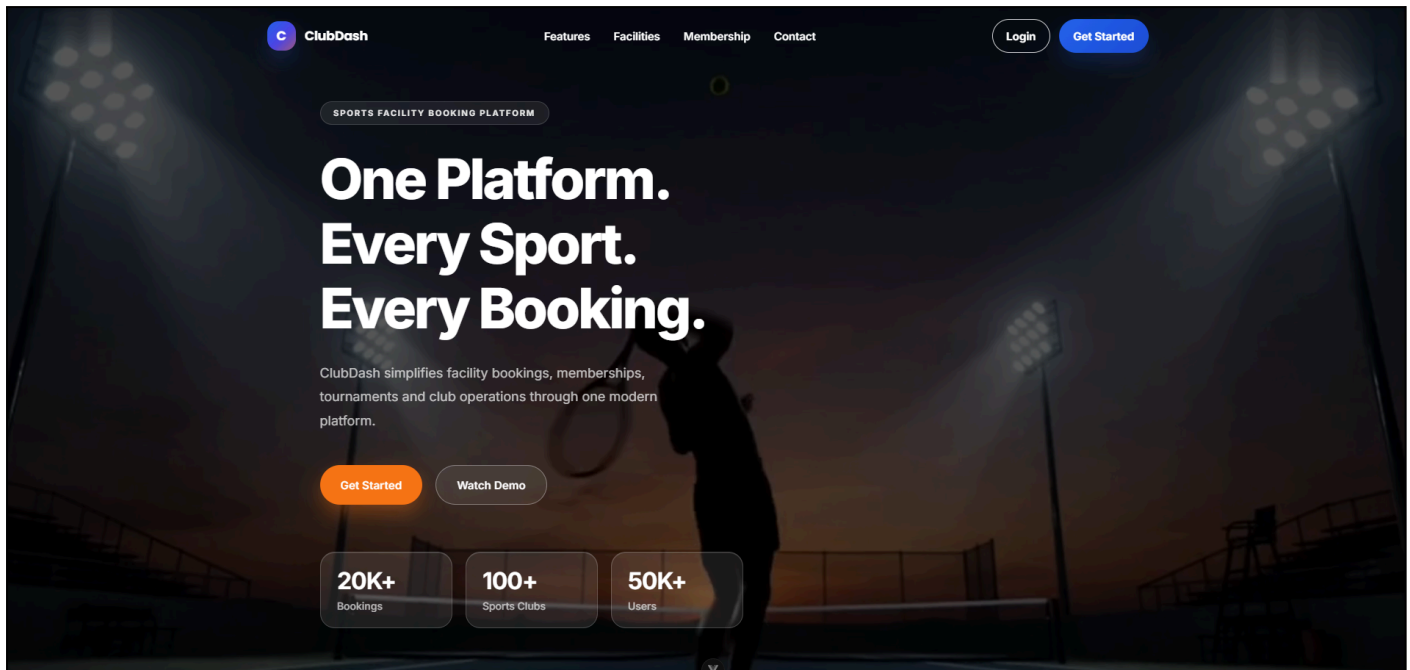
Features

- Responsive navigation bar with Login and Register options.
- Hero section with sports-themed background video.
- Introduction to the Sports Facility Booking & Club Management System.
- Facilities section showcasing available sports amenities.
- Membership section highlighting membership benefits.
- Events preview section displaying upcoming activities.
- Newsletter subscription section.
- Footer with navigation links and contact information.
- Fully responsive layout with smooth scrolling and hover animations.

Purpose

The landing page provides visitors with an overview of the platform while encouraging new users to register and existing users to log in. It also showcases the club's facilities, memberships, and events in an engaging and user-friendly manner.

Figure 1. Landing Page Interface



2. Login Page (/login)

The Login Page enables registered users to securely access the Sports Facility Booking & Club Management System using their credentials. The interface is designed to be simple, intuitive, and responsive across different screen sizes.

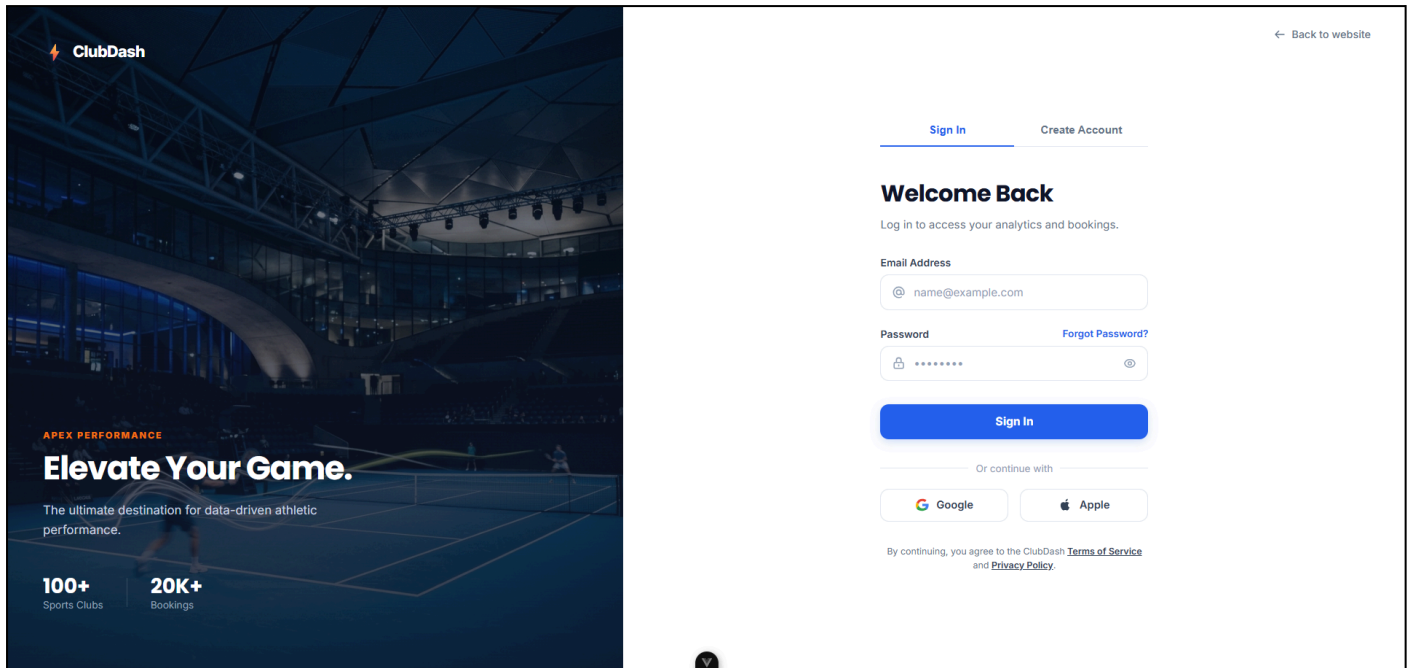
Features

- Email and password input fields.
- Secure password entry.
- Login button for authentication.
- Navigation link to the registration page.
- Form validation for required fields.
- Responsive card-based design with modern styling.

Purpose

The login interface provides secure authentication for existing users and serves as the gateway to the personalized dashboard and booking features.

Figure 2. Login Page Interface



3. Registration Page (/register)

The Registration Page allows new users to create an account and gain access to the Sports Facility Booking & Club Management System. The page collects essential user information while maintaining a clean and responsive interface.

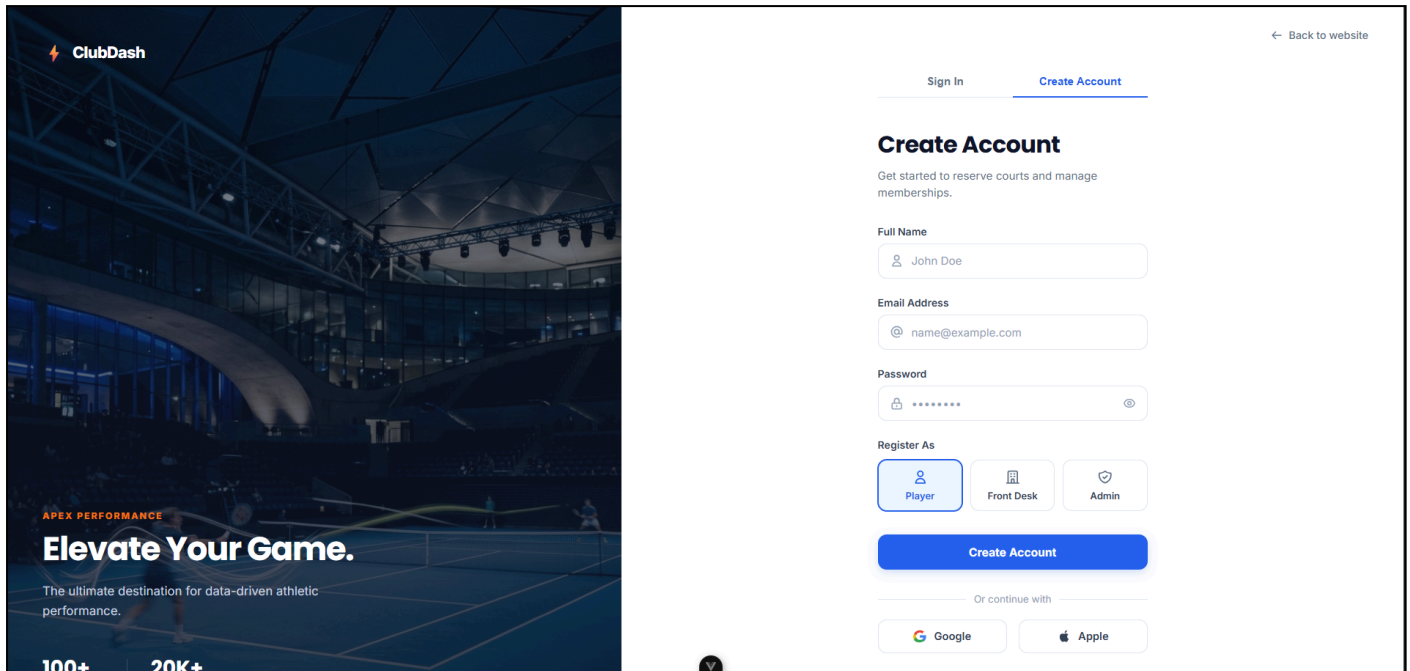
Features

- User registration form.
- Name, email, password, and confirm password fields.
- Client-side validation.
- Register button for account creation.
- Link to the login page.
- Responsive layout optimized for mobile and desktop devices.

Purpose

The registration page simplifies the onboarding process by allowing new users to quickly create an account and access the platform's booking and club management features.

Figure 3. Registration Page Interface



4. User Dashboard (/dashboard)

The **User Dashboard** serves as the primary landing page after login. It provides users with a personalized overview of their activities and quick access to commonly used features.

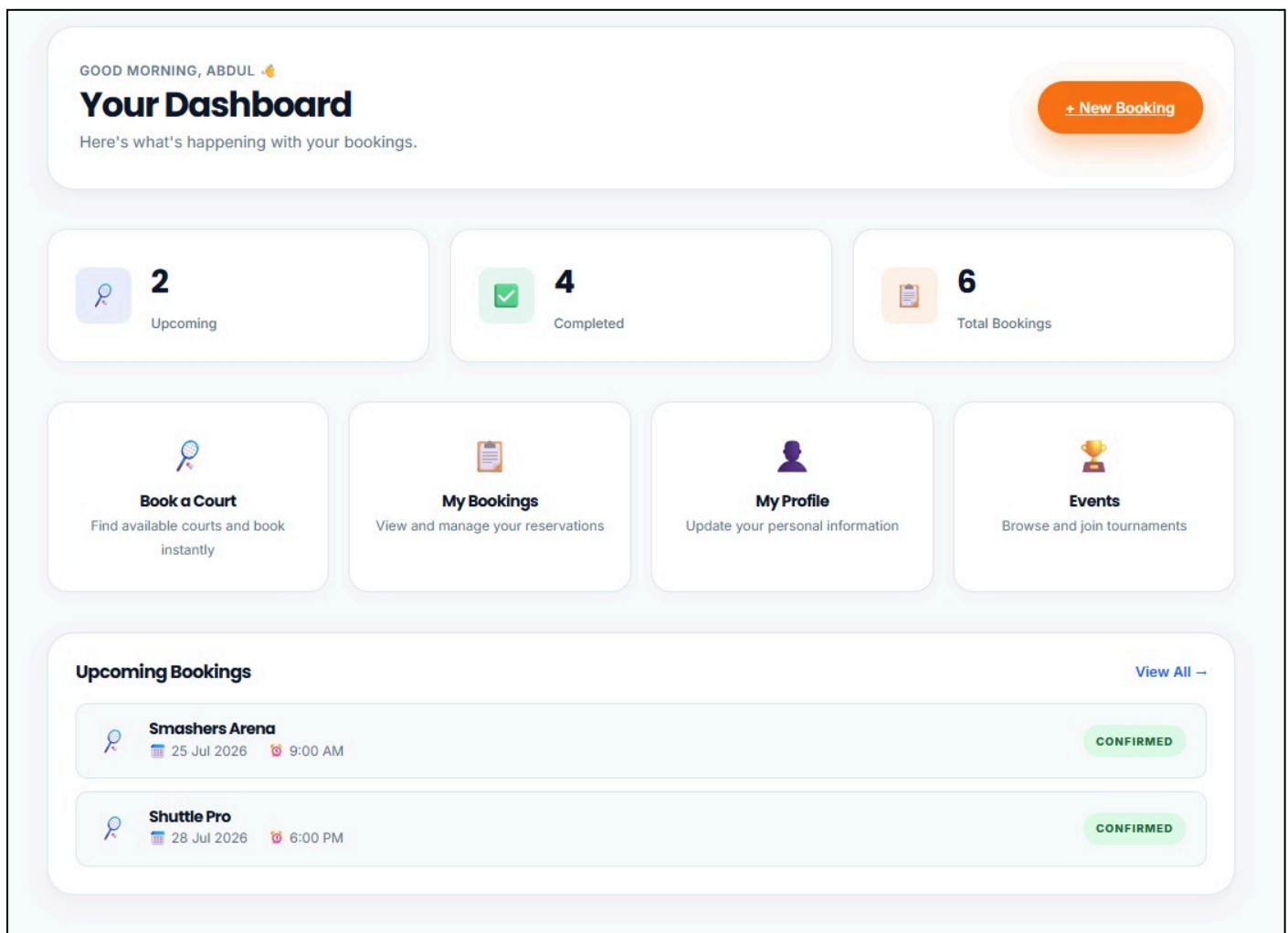
Features

- Dynamic greeting based on the current time of day.
- Statistics cards displaying Upcoming Bookings, Completed Bookings, and Total Bookings.
- Quick action buttons for:
 - Book a Court
 - My Bookings
 - Profile
 - Events
- Upcoming bookings section displaying the court name, booking date, and time.
- "View All" navigation link for accessing the complete booking history.
- Fully responsive layout with modern card design and hover animations.

Purpose

The dashboard enables users to quickly monitor their booking activity and navigate efficiently to the most frequently used sections of the system.

Figure 4. User Dashboard Interface



5. Booking Dashboard (/booking)

The **Booking Dashboard** is the primary reservation interface where users can browse available sports courts and make bookings efficiently.

Features

- Date picker with today's date selected by default.
- Time slot filters:

- Morning
 - Afternoon
 - Evening
- Court cards displaying:
 - Court name
 - Availability status
 - Booking price
- Status badges indicating Available, Booked, or Maintenance.
- Disabled booking buttons for unavailable courts.
- Responsive grid layout:
 - Desktop: 3 columns
 - Tablet: 2 columns
 - Mobile: 1 column
- Reusable **CourtCard** component for consistent presentation.

Purpose

The booking interface allows users to easily identify available courts, compare pricing, and reserve sports facilities while preventing invalid bookings through clear availability indicators.

Figure 5. Booking Dashboard Interface

COURT BOOKING

Find and Reserve Your Court

Select your preferred court, date, and time slot below.

SELECT DATE
23-07-2026

TIME SLOT
Morning Afternoon Evening

Court Name	Court ID	Rate	Status	Action
Smashers Arena	Court 1	₹450 / hour	AVAILABLE	Select & Book
Net Ninja Court	Court 2	₹500 / hour	BOOKED	Unavailable
Lightning Strike	Court 3	₹400 / hour	AVAILABLE	Select & Book
Ace Badminton	Court 4	₹350 / hour	MAINTENANCE	Unavailable
Shuttle Pro	Court 5	₹600 / hour	AVAILABLE	Select & Book
Champions Court	Court 6	₹550 / hour	BOOKED	Unavailable

6. My Bookings (/bookings)

The **My Bookings** page enables users to view, organize, and manage all their court reservations from a single interface.

Features

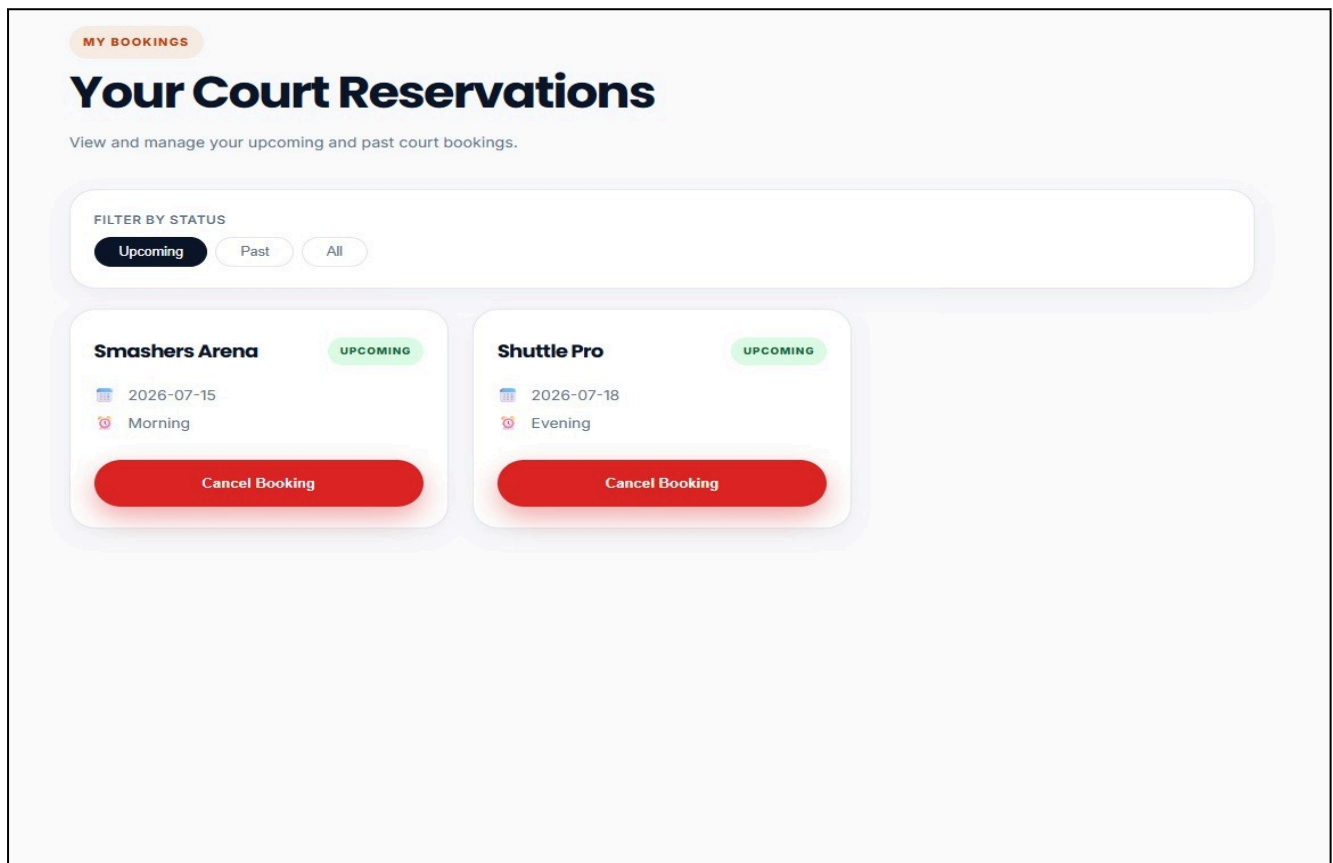
- Tab-based booking filters:
 - Upcoming
 - Past
 - All
- Card-based layout displaying:
 - Court name
 - Booking date
 - Time slot
 - Booking status
- Cancel booking option for upcoming reservations.

- Color-coded booking status indicators.
- Empty-state message with a call-to-action for users who have no bookings.

Purpose

This page provides users with a centralized location to monitor their reservation history, manage future bookings, and improve the overall booking experience.

Figure 6. My Bookings Interface



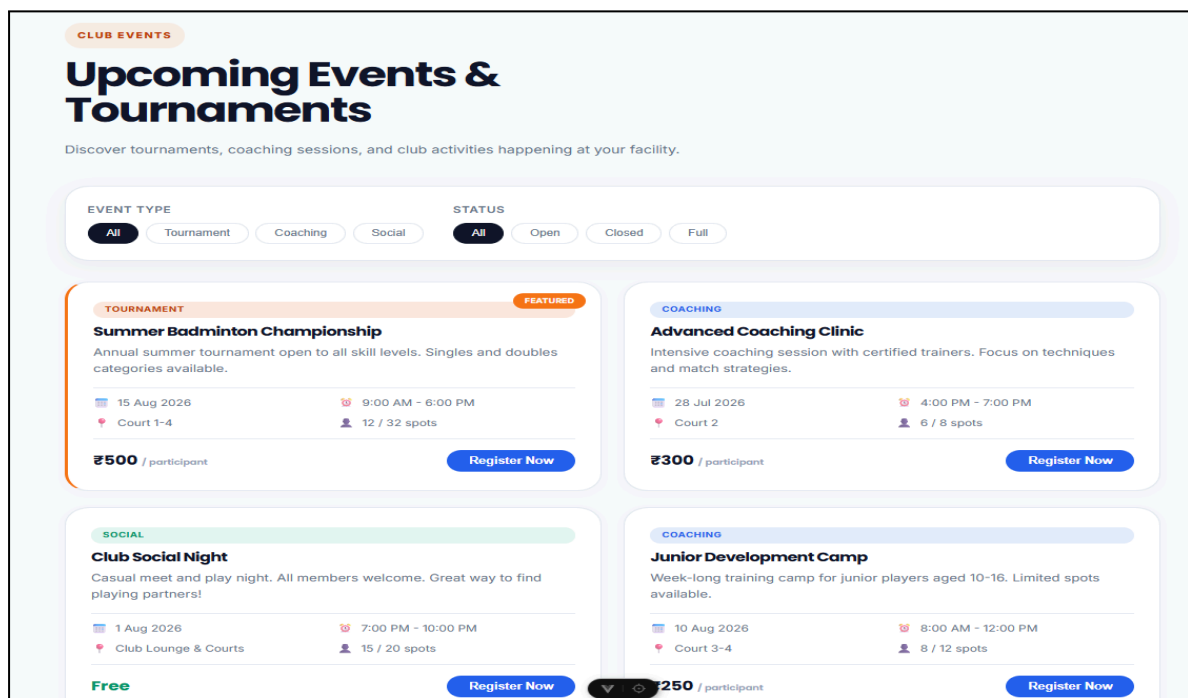
7. Events Page (/events)

The Events Page allows users to discover and register for club events, tournaments, and coaching sessions. It provides a comprehensive view of all upcoming activities at the facility with filtering and registration capabilities.

Features:

- Event cards displaying title, description, date, time, and location
- Type filters (All, Tournament, Coaching, Social)
- Status filters (All, Open, Closed, Full)
- Participant count with maximum spots display
- Pricing display (free or paid events)
- Register button with status-based states (Open, Closed, Full)
- Featured events with visual highlight
- Empty state for no events
- Fully responsive layout

Figure 7: Events Page Interface



Summary

The frontend interfaces were developed using **Vue.js 3** and designed with a strong emphasis on **usability, responsiveness, and consistency**. By adopting reusable components, a centralized styling system, and a mobile-first responsive layout, the application delivers a clean and intuitive user experience across desktops, tablets, and mobile devices while effectively supporting the sports facility booking workflow.

8. Admin Dashboard (/admin)

The **Admin Dashboard** serves as the central management interface for administrators of the Sports Facility Booking & Club Management System. It provides a comprehensive overview of club operations, including member statistics, court availability, bookings, events, and facility analytics. The dashboard is designed with a modern card-based layout, enabling administrators to efficiently monitor key metrics and perform day-to-day management tasks.

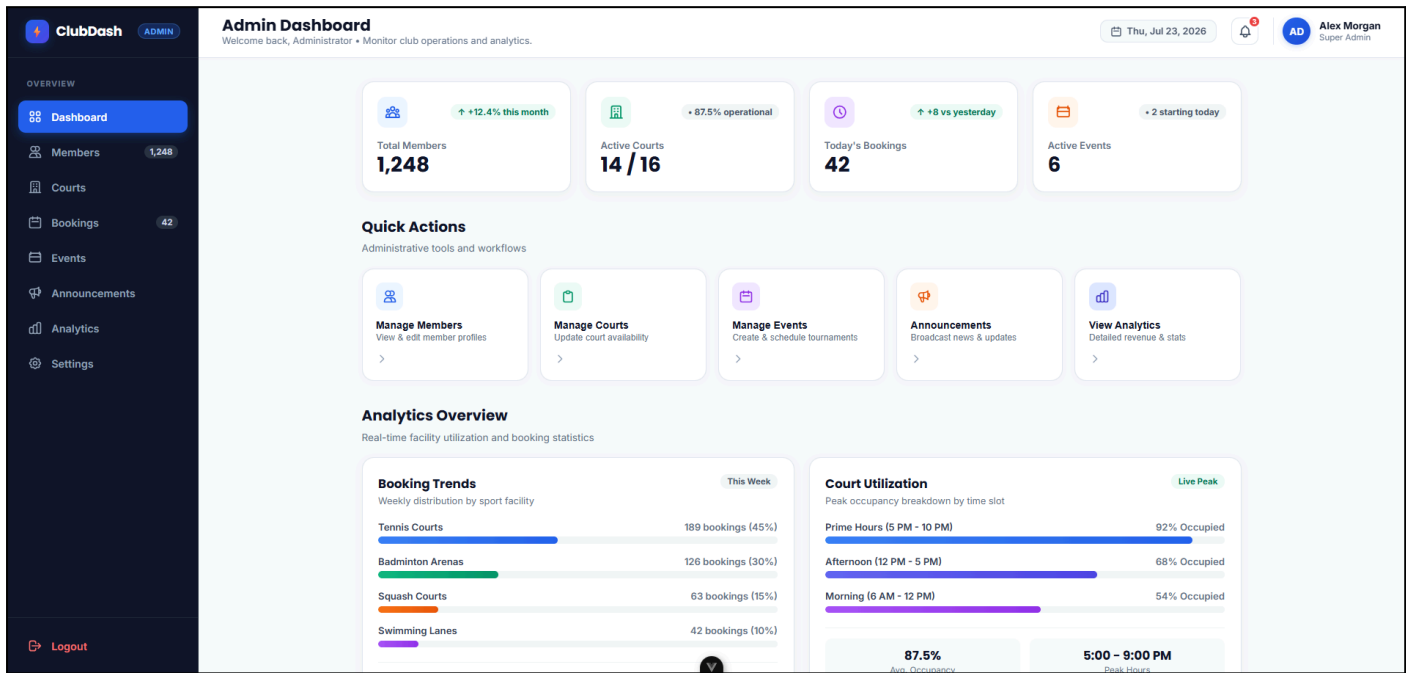
Features

- Administrative overview displaying Total Members, Active Courts, Today's Bookings, and Active Events.
- Sidebar navigation for quick access to Dashboard, Members, Courts, Bookings, Events, Announcements, Analytics, and Settings.
- Quick Action cards for:
 - Manage Members
 - Manage Courts
 - Manage Events
 - Announcements
 - View Analytics
- Analytics Overview section displaying booking trends across different sports facilities.
- Court Utilization statistics showing occupancy rates across different time slots.
- Notification icon and administrator profile section.
- Responsive dashboard layout with reusable cards, progress indicators, and hover animations.

Purpose

The Admin Dashboard provides club administrators with a centralized interface to monitor facility operations, manage resources, analyze booking trends, and efficiently perform administrative tasks through a clean and intuitive user interface.

Figure 8. Admin Dashboard Interface



9. Members Management (/admin)

Overview

club members and membership requests. It enables efficient monitoring of member statistics, review of pending applications, and management of membership approvals through a clean and organized interface. The **Members Management** interface provides administrators with a centralized view of all registered

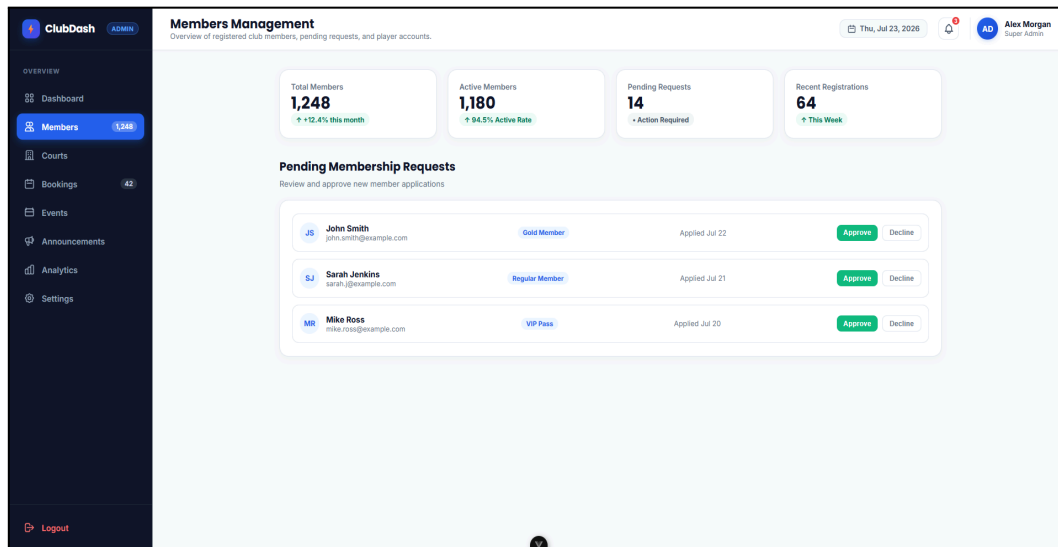
Features

- Summary cards displaying Total Members, Active Members, Pending Requests, and Recent Registrations.
- Pending Membership Requests section showing new membership applications.
- Member cards displaying applicant name, email address, membership type, and application date.
- Membership categories such as Gold Member, Regular Member, and VIP Pass.
- Approve and Decline action buttons for processing membership requests.
- Responsive card-based layout with reusable UI components and status badges.
- Sidebar navigation for quick access to other administrative modules.

Purpose

The Members Management interface streamlines the membership approval process by providing administrators with a centralized platform to review applications, monitor membership statistics, and efficiently manage player accounts while reducing manual administrative effort.

Figure 9. Members Management Interface



10. Bookings & Reservations (/admin)

Overview

The **Bookings & Reservations** interface provides administrators with a centralized view of all court reservations and booking activities. It enables efficient monitoring of active bookings, upcoming reservations, completed sessions, and cancellations, allowing administrators to oversee court scheduling and ensure smooth facility operations.

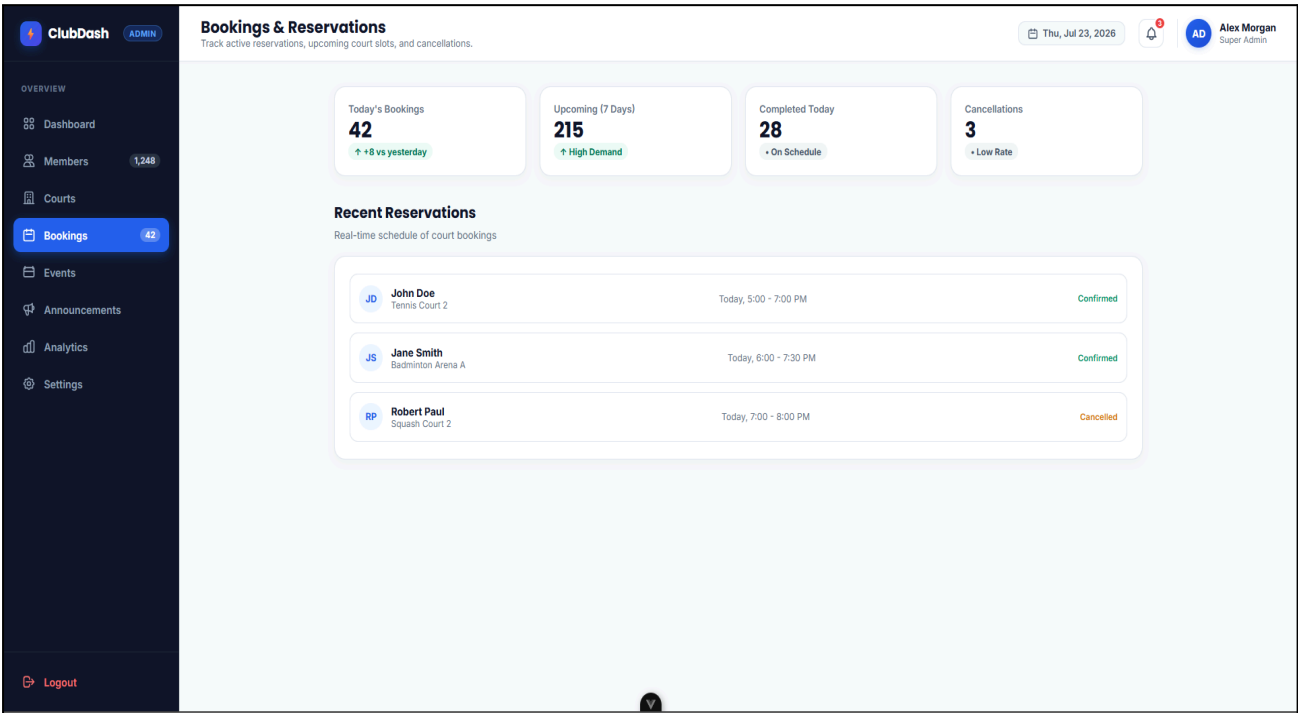
Features

- Summary cards displaying Today's Bookings, Upcoming Reservations, Completed Bookings, and Cancellations.
- Recent Reservations section presenting the latest court bookings in real time.
- Reservation entries displaying:
 - Member name
 - Reserved court
 - Booking date and time
 - Reservation status
- Color-coded booking status indicators such as **Confirmed** and **Cancelled**.
- Real-time booking overview for monitoring daily court schedules.
- Responsive card-based layout with reusable UI components and consistent styling.
- Sidebar navigation for seamless access to other administrative modules.

Purpose

The Bookings & Reservations interface enables administrators to efficiently track reservation activities, monitor booking schedules, and oversee cancellations from a single dashboard. By providing a real-time overview of booking information, the system helps minimize scheduling conflicts and improves overall booking management.

Figure 10. Bookings & Reservations Interface



11. Performance & Analytics (/admin)

Overview

The **Performance & Analytics** interface provides administrators with comprehensive insights into the club's operational performance through real-time statistics and visual analytics. It enables monitoring of revenue, booking trends, facility utilization, and membership growth, supporting data-driven decision-making for efficient club management.

Features

- Summary cards displaying Total Revenue, Peak Booking Hour, Membership Growth, and Member Retention Rate.

- Booking Trends section illustrating reservation distribution across different sports facilities.
- Peak Hours Occupancy dashboard showing court utilization during various time slots.
- Visual progress indicators for analyzing booking patterns and facility usage.
- Real-time occupancy metrics for monitoring peak and off-peak hours.
- Responsive analytics dashboard with reusable cards and graphical progress components.
- Sidebar navigation provides quick access to other administrative modules.

Purpose

The Performance & Analytics interface enables administrators to evaluate booking patterns, monitor facility utilization, track membership growth, and analyze operational performance through centralized visual reports. These insights support informed decision-making, improve resource allocation, and enhance the overall efficiency of sports club management.

Figure 11. Performance & Analytics Interface

